

AL AQIQ, Saudi Arabia

WMO: 410550

Lat: 20.296N

Lon: 41.634E

Elev: 1672

StdP: 82.79

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.8	9.0	-9.2	2.1	18.1	-7.1	2.5	20.2	8.4	19.3	7.2	19.6	1.9	120	0.388

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.2	36.1	16.8	35.6	16.8	35.0	16.7	21.2	28.9	20.4	28.6	19.7	28.4	5.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	16.9	26.3	17.9	15.8	24.9	17.0	14.9	23.9	70.6	29.4	67.2	28.4	64.5	28.3	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.1	7.1	DB	4.9	37.5	1.0	0.7	4.2	38.0	3.6	38.4	3.0	38.7	2.3	39.2	
			WB	0.8	22.5	0.8	0.7	0.2	23.0	-0.2	23.4	-0.7	23.8	-1.2	24.3	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.6	16.8	18.8	21.0	23.3	26.5	29.2	29.5	27.8	23.4	19.7	17.1
		DBStd	5.03	2.56	2.50	2.28	2.22	1.99	1.16	1.34	1.05	1.37	1.54	1.84
		HDD10.0	1	1	0	0	0	0	0	0	0	0	0	0
		HDD18.3	144	59	22	6	2	0	0	0	0	0	8	48
		CDD10.0	4960	212	248	342	401	511	576	604	534	417	291	221
		CDD18.3	2062	12	36	89	152	252	326	346	284	159	49	11
		CDH23.3	23342	86	253	609	1188	2603	4286	4588	4548	3424	1351	325
(13)		CDH26.7	11039	6	31	118	387	1188	2293	2497	2451	1663	373	31
(14)	Wind	WSAvg	3.2	2.5	2.8	2.9	2.8	4.0	4.7	4.4	3.4	2.8	2.4	2.3
(15)	Precipitation	PrecAvg	132	12	1	11	37	23	6	10	9	3	11	8
		PrecMax	283	109	3	64	131	68	23	106	25	16	56	43
		PrecMin	39	0	0	0	1	0	0	0	0	0	0	0
		PrecStd	60	23	1	16	41	17	6	20	8	4	16	11
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	27.2	29.1	30.8	32.6	35.1	36.5	37.1	36.9	35.7	31.3	28.8
			MCWB	13.9	14.5	15.8	15.6	16.4	16.8	17.3	17.2	17.1	14.7	15.3
		2%	DB	25.4	27.8	29.1	31.2	34.1	35.9	36.2	36.0	34.5	30.4	27.7
			MCWB	13.8	14.8	15.6	16.0	16.1	16.5	16.9	16.8	16.5	14.2	14.8
		5%	DB	24.0	26.2	28.0	30.1	33.1	35.1	35.8	35.2	33.9	29.8	26.8
			MCWB	14.0	14.6	15.4	16.6	16.2	16.3	16.9	17.0	16.1	14.1	14.8
		10%	DB	22.6	24.9	26.9	29.0	32.1	34.2	34.8	34.8	33.0	28.9	25.2
			MCWB	13.7	14.6	15.7	16.5	16.4	16.1	16.8	17.0	15.8	13.9	14.6
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.1	18.5	19.7	20.6	20.8	21.1	22.3	22.4	22.0	19.2	18.7
			MCDB	23.0	24.3	26.6	27.6	28.6	29.9	30.6	29.4	30.0	24.2	24.0
		2%	WB	16.2	17.2	18.6	19.6	19.8	19.7	20.9	21.3	19.6	18.3	17.9
			MCDB	21.9	23.5	25.2	26.3	27.6	30.3	29.4	28.5	28.4	24.3	23.8
		5%	WB	15.5	16.5	17.9	18.8	19.0	18.8	19.9	20.4	18.4	17.4	17.1
			MCDB	21.1	22.8	24.4	25.4	26.9	30.1	29.5	28.5	28.9	24.0	22.9
		10%	WB	14.9	15.7	17.1	18.1	18.3	18.1	19.0	19.5	17.5	16.3	16.4
			MCDB	20.5	22.1	23.7	24.9	26.9	30.1	29.9	29.0	29.5	24.3	22.2
(35)	Mean Daily Temperature Range	MDBR	11.7	12.0	11.6	11.3	11.5	11.4	11.2	11.2	10.9	11.2	11.6	11.8
		5% DB	MCDBR	13.0	13.1	12.4	12.1	11.8	11.7	11.6	11.3	11.4	12.5	12.7
			MCWBR	6.8	6.9	6.2	5.5	5.1	5.5	5.4	5.1	5.1	5.9	6.6
		5% WB	MCDBR	11.4	11.9	10.8	11.2	11.5	11.2	11.2	10.7	11.1	11.3	11.0
			MCWBR	6.8	7.1	5.7	5.3	5.1	6.0	6.1	5.5	5.6	6.0	6.4
(40)	Clear-Sky Solar Irradiance	taub	0.342	0.370	0.408	0.454	0.514	0.546	0.579	0.593	0.512	0.404	0.373	0.344
		taud	2.390	2.273	2.146	2.029	1.851	1.755	1.717	1.684	1.833	2.134	2.267	2.372
		Ebn at Noon	933	923	898	857	798	766	742	735	792	875	889	917
		Edh at Noon	111	132	155	177	210	229	238	247	210	149	124	109
(44)	All-Sky Solar Radiation	RadAvg	4.78	5.63	6.45	6.82	7.27	7.52	7.02	6.62	6.50	6.17	5.13	4.63
		RadStd	0.31	0.30	0.29	0.36	0.34	0.21	0.25	0.18	0.19	0.20	0.22	0.31

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	-35	+154	+127

Nomenclature: See separate page